

OurScheme – Government Scheme Finder

Complete Technical Documentation, File-by-File Architecture & API Guide

Project: OurScheme Government Portal	Architecture: Flask MVC + MySQL 8.0 + Google GenAI + Translate
Categories: Exactly 16 Validated Categories	Schemes Catalog: 100 Government Welfare Schemes
Key Features: AI & Rule Eligibility, Multilingual, Admin Panel	Admin URL: http://localhost:5000/admin (admin / admin123)

1. Project Executive Summary

OurScheme is an end-to-end, citizen-centric government scheme aggregation and eligibility discovery platform. Citizens in India frequently miss out on central and state welfare programs due to fragmented information, complex criteria, and language barriers. OurScheme resolves this by consolidating over 100 welfare programs across **16 distinct socioeconomic categories**, delivering instant personalized eligibility evaluation through both a structured 5-step demographic wizard and an AI-powered conversational query engine, providing live multi-language accessibility across 10 Indian languages, and providing a secure administrative portal for scheme governance.

Core Functionalities:

- **Categorized Scheme Browsing:** Organized into 16 verified categories (Agriculture, Education, Healthcare, Social Welfare, Women & Child Development, Employment, Housing, Financial Services, etc.) with scheme counts and details.
- **5-Step Deterministic Eligibility Checker:** Analyzes age, gender, caste (General/OBC/SC/ST), employment, income ceiling, state, and house ownership to match qualifying schemes with pass/fail reasons.
- **Conversational AI Profile Extraction:** Citizens can describe their situation in natural language (e.g., 'I am a 21yo female student from Maharashtra with family income under 1.5 lakhs'); Google GenAI (Gemini) extracts structured parameters with offline regex fallback.
- **Dual-Layer Multilingual Architecture:** Instant client-side DOM translation for essential navigation and buttons paired with Google Website Translation Element API for deep page translation across 10 Indian languages (Hindi, Marathi, Gujarati, Bengali, Tamil, Telugu, Kannada, Malayalam, Punjabi, English).
- **Citizen Account Management:** Secure user registration, authentication, and a personal 'Saved Schemes' bookmarking dashboard.
- **Comprehensive Admin Dashboard:** Located at /admin with summary metric cards (Total Schemes, Total Categories, Total Contact Messages), full Scheme CRUD (Add/Edit/Delete), category monitoring, and contact inquiry message management.

2. APIs and Integrations Used

The project integrates external cloud APIs, client-side translation APIs, database communication layers, and internal cryptographic frameworks as detailed below:

API / Integration	Type / Source	Purpose & Implementation Details	Security & Resilience
Google Website Translation Element API	Client-Side API Google LLC	Dynamically translates DOM content across 10 Indian languages via <code>//translate.google.com/translate_a/element.js?cb=googleTranslateElementInit</code> . Synchronized with active language select dropdown in navbar.	Persists language in googtrans cookie. Paired with instant client dictionary for immediate UI feedback.
Google GenAI API (Gemini 2.5 Flash)	Cloud AI SDK Google DeepMind	Used in <code>ai_extractor.py</code> to interpret freeform conversational text into structured demographic parameters (age, income, category, gender, state). Also used for backend translations in <code>translator.py</code> .	Built-in offline regex NLP fallback parser guarantees 100% uptime if <code>GEMINI_API_KEY</code> is unset or offline.
MySQL Connector Python API	Database Driver Oracle MySQL	Handles TCP connections to MySQL database <code>government_schemes</code> . Executes parameterized SQL statements for public scheme discovery, user authentication, and admin CRUD operations.	Strict parameterized statements (<code>%s</code> placeholders) preventing SQL injection attacks.

Werkzeug Security API	Crypto Library Pallets Projects	Manages one-way cryptographic password hashing for citizen accounts (users) and administrative accounts (admins) using generate_password_hash and check_password_hash.	Salted PBKDF2/scrypt hashes prevent plain-text exposure even if database is compromised.
Flask Session & Context API	Internal Web API Flask Framework	Manages authenticated sessions for citizens (user_id) and admins (admin_logged_in). Context processor injects active language state into every rendered Jinja2 template.	Cryptographically signed client-side session cookies using server SECRET_KEY.

3. Purpose of Each File in the Project

File Path	Role	Detailed Purpose and Responsibilities
<code>app.py</code>	Core Controller	Central application hub. Configures Flask app, database connection helper, session lifecycle, error handlers, and context processors. Defines all routes for scheme browsing, search, user authentication, 5-step eligibility assessment, AI query processing, bookmarking, multilingual sync (/set_language), and the entire /admin control suite.
<code>eligibility_engine.py</code>	Business Logic	Deterministic rules engine. Compares user profiles against database criteria (age range, gender, caste, employment, max income ceiling, state, house ownership). Calculates match percentages and generates human-readable explanations of why the citizen is eligible.
<code>ai_extractor.py</code>	AI / NLP Module	Extracts structured demographic attributes from natural language queries using Google GenAI (Gemini 2.5 Flash). Includes an integrated offline regex extractor to ensure offline resilience for age, income (LPA, k, lakhs), gender, and caste.
<code>translator.py</code>	Localization	Defines dictionary of 10 supported Indian languages (code, English name, native script). Provides backend translation caching and fallback translation functions using Google GenAI.
<code>create_admin_table.py</code>	DB Setup	Creates the admins table in MySQL with secure hashed password storage. Seeds the default super-administrator account (admin / admin123).
<code>run_migrations.py</code>	DB Migration	Schema migration runner: ensures schemes table has criteria columns, creates users and saved_schemes tables, cleans and normalizes categories to exactly 16 unique entries, and sets realistic eligibility criteria for 100 schemes.
<code>migrations.sql</code>	SQL Script	Raw SQL migration containing DDL statements for creating tables and DML updates for criteria seeding.
<code>test_complete_flow.py</code>	Test Suite	Automated integration test verifying all public citizen journeys: homepage, category loading, search queries, registration/login, 5-step wizard, AI extraction, and scheme saving.
<code>test_admin_suite.py</code>	Test Suite	Automated test suite verifying the Admin Dashboard: session security, login/logout, metric card calculations, scheme addition, editing, deletion, and contact message management.
<code>static/style.css</code>	Frontend CSS	Global stylesheet for public website: navbar, hero carousel, category cards, buttons, footer, responsive grids, and cleanly hiding Google Translate default top banners.
<code>static/admin.css</code>	Admin CSS	Dedicated styles for the Admin Panel: dark sidebar, modern stat cards, structured data tables, status badges, modal dialogs, and clean admin login UI.
<code>static/eligibility.css</code>	Frontend CSS	Styles the 5-step interactive eligibility popup modal, progress bar, radio badges, step indicators, and AI prompt input tab.
<code>static/eligibility.js</code>	Frontend JS	Client-side controller for the eligibility wizard: step navigation, form validation, AJAX submission to /check_eligibility, and conversational AI prompt handling.
<code>static/language.js</code>	Frontend JS	Dual-layer language switcher: provides instant client dictionary translation for UI navigation and triggers Google Website Translation Element API with cookie synchronization.
<code>static/test.css</code>	Frontend CSS	Styles scheme catalog cards, category tags, eligibility criteria pill badges, and bookmark icons in test.html.
<code>static/ContactUs.css</code>	Frontend CSS	Styles the Contact Us inquiry page: office information cards, form layout, and status alerts.
<code>static/AboutUs.css</code>	Frontend CSS	Styles the About Us page, mission statement, platform benefits, and team vision cards.
<code>templates/base.html</code>	Base Template	Master public HTML layout containing top navigation, search bar, language switcher dropdown, citizen auth buttons, flash alert containers, and footer.
<code>templates/home.html</code>	Public HTML	Landing homepage featuring hero banner carousel, quick CTA for eligibility check, and dynamic cards for the 16 government scheme categories.
<code>templates/test.html</code>	Public HTML	Scheme catalog listing displaying schemes in cards with category badges, description, eligibility criteria, and 'Save Scheme' bookmarking.
<code>templates/eligibility.html</code>	Public HTML	Interactive 5-step eligibility modal questionnaire and conversational AI prompt tab.

templates/eligibility_results.html	Public HTML	Results page displaying matched qualifying schemes, eligibility match percentages, specific qualification reasons, and application links.
templates/login.html	Public HTML	Citizen authentication login form with email/password validation.
templates/register.html	Public HTML	Citizen registration form capturing account credentials and optional demographic profile.
templates/saved_schemes.html	Public HTML	Personalized bookmark dashboard where authenticated citizens access their saved schemes.
templates/contactUs.html	Public HTML	Citizen feedback and inquiry submission form connecting directly to the database.
templates/AboutUs.html	Public HTML	Informational page describing platform objectives, background, and government initiatives.
templates/admin/admin_base.html	Admin Base	Master layout for the Admin Panel: sidebar navigation, active page indicator, admin header, and secure logout link.
templates/admin/admin_login.html	Admin HTML	Dedicated login page for portal administrators at /admin/login.
templates/admin/dashboard.html	Admin HTML	Admin overview dashboard displaying 3 real-time summary KPI cards (Total Schemes, Total Categories, Total Contact Messages) and quick links.
templates/admin/schemes.html	Admin HTML	Comprehensive scheme data table with live search, category filter, and edit/delete actions.
templates/admin/add_scheme.html	Admin HTML	Form for administrators to create new welfare schemes with category selection and criteria.
templates/admin/edit_scheme.html	Admin HTML	Form pre-populated with existing scheme details allowing updates to criteria and information.
templates/admin/categories.html	Admin HTML	Category overview table presenting the 16 active categories and scheme counts per category.
templates/admin/contacts.html	Admin HTML	Message inbox displaying citizen inquiries from Contact Us with timestamp and delete actions.

4. Database Architecture & Table Structure

Table Name	Primary Key	Columns & Datatypes	Purpose & Constraints
schemes	id (INT AUTO)	scheme_name, category, description, min_age, max_age, gender, caste_category, employment_status, max_income, house_owner, state	Houses 100 government welfare schemes with structured eligibility criteria.
categories	id (INT AUTO)	category_name (VARCHAR 100), description, icon_path	Exactly 16 distinct categories organizing schemes.
users	id (INT AUTO)	username, email, password_hash, gender, age, caste, state, income	Stores citizen profiles and credentials with secure hashing.
admins	id (INT AUTO)	username (VARCHAR 50 UNIQUE), password_hash (VARCHAR 255), email, created_at	Administrative accounts managing portal access at /admin.
contacts	id (INT AUTO)	name, email, phone, message, submitted_at	Stores citizen feedback and contact inquiries.
saved_schemes	id (INT AUTO)	user_id (INT), scheme_id (INT), saved_at	Junction table managing citizen bookmarks with unique compound constraint.

5. Technical Verification & Execution Instructions

- **Run Flask Server:** Execute python app.py in the workspace. Access the portal at http://127.0.0.1:5000.
- **Access Admin Dashboard:** Visit http://localhost:5000/admin. Sign in with username: admin and password: admin123.
- **Run End-to-End Test Suite:** Run python test_complete_flow.py to verify citizen journeys.
- **Run Admin Test Suite:** Run python test_admin_suite.py to verify admin CRUD operations and security authorization.
- **Multilingual Functionality:** Use the top navbar language dropdown to switch among 10 languages (Hindi, Marathi, Gujarati, etc.).
- **AI Eligibility Evaluation:** Enter natural language text into the eligibility modal AI tab. The system extracts demographic parameters using Google GenAI or offline regex fallback.